

AKRITI DHASMANA

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EDUCATION

University of Colorado, Boulder

Ph.D. in Computer Science and Engineering

Advisors: Dr. Maria Antoniak, Dr. Alexis Palmer

Boulder, CO

2026 – 2031

University of Notre Dame

M.Sc. in Computer Science and Engineering

Thesis: Automated Speech Recognition for Indic Dialects and Language Varieties

Thesis Defense Committee: Dr. David Chiang, Dr. Meng Jiang, Dr. Xiangling Zhang

South Bend, IN

2026

Union College

B.Sc. in Computer Science and English, Minor in Mathematics

Schenectady, NY

2022

RESEARCH & EXPERIENCE

University of Colorado, Boulder

Research Assistant

2026 – 2031

Boulder, CO

- Researching narrative understanding and low-resource languages with Dr. Maria Antoniak and Dr. Alexis Palmer.

University of Notre Dame

Research Thesis Student

2025 – 2026

South Bend, IN

- Finetuned ASR models (XLS-R, HuBERT, wav2vec2/BERT) on low-resource Indic dialects under Dr. David Chiang. Developed cross-lingual transfer learning donor selection methods based on linguistic distance metrics.

Chartbeat

Data Engineer II

2022 – 2025

New York, NY

- Built recommendation engine and maintained worker-queue architecture handling up to 1B API requests daily.
- Integrated Google Search Console SEO metrics and deployed Python/C++/Java/Lua microservices on Kubernetes across Redshift, Postgres, and Snowflake.

Union College

Scholar Research Student / Research Assistant

2019 – 2021

Schenectady, NY

- Mapped fictional character power dynamics using Relational Character Analysis Tool and evaluated coreference tools (NeuralCoref, AllenNLP) on Hinglish texts guided by Dr. Kristina Striegnitz.
- Implemented Permutation Decision Diagrams for matrix multiplication research. Designed and benchmarked a novel Python data structure for cyclic abelian abstract groups under the guidance of Dr. Matt Anderson.

International Business Machines (IBM)

Summer Research Intern

Summer 2020

Yorktown Heights, NY

- Assisted Dr. Wendy Cornell on automated literature text-mining across arXiv, ChemRxiv, PubChem, and ChEMBL for COVID-19 drug discovery

PUBLICATIONS

- **Dialect Matters: Cross-Lingual ASR Transfer for Low-Resource Indic Language Varieties** 2026
Authors: Akriti Dhasmana, Aarohi Srivastava, and David Chiang.
In Proceedings of the 13th Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial), pp. 145–156. Association for Computational Linguistics. Rabat, Morocco.

POSTER PRESENTATIONS

Dialect Matters: Cross-Lingual ASR Transfer for Indic Languages – MSLD	2026
The Billion-Request Content Recommendation System Challenge – PyData Conference, NYC	2023
Hierarchical SNA of Short Fictional Texts – Steinmetz Symposium, Union College	2022
Group Decision Diagrams (GDDs) – Steinmetz Symposium, ACM NYCWIC	2021
πDD: Permutation Decision Diagrams – New York Six Research Consortium	2019

TEACHING & MENTORSHIP

University of Colorado Boulder – Mentor, CS Directed Reading Program Mentor an undergraduate STEM student, assigned weekly readings, and led academic discussion sessions.	2026
University of Notre Dame – Teaching Assistant & AWIS Mentor Supported Discrete Math & Data Structures courses; mentored STEM undergraduate students.	2025 – 2026
WeSpeakNYC – English Language Class Facilitator Led weekly conversational English classes for adult immigrants, focusing on civic engagement.	2022 – 2024
Union College – TA, Orientation Leader & Mentor TA for Game Development; served as Orientation Advisor, and Mensa Meaning Making Mentor.	2021 – 2024

LEADERSHIP & SERVICE

Chartbeat DEI Committee (Member) – Coordinated mentorship events and a book club.	2022 – 2024
Union College ACM-Women (President) – Led workshops, faculty events, and conference trips.	2019 – 2022
Concordiensis (Co-Editor-in-Chief) – Supervised staff and led weekly editorials.	2020 – 2021
Union College Robotics Crew (Programming Leader)	2021 – 2022

AWARDS & FELLOWSHIPS

CU Boulder: Awtar and Teji Singh Graduate Fellowship [1st-year CS/Civil student]; CEAS LSAMP Bridge Fellow [Mentoring, networking, and travel funding for underrepresented STEM grads].
Union College: Phi Beta Kappa [Academic honor society]; Sigma Xi [Scientific research honor society]; Alice P. & Donald C. Loughry CS Senior Project Prize [Outstanding senior project]; CS Department Service Prize; Omicron Delta Kappa [Leadership honor society]; Sigma Tau Delta [English honor society]; Most Impactful Leader in Media Prize; Eugene I. Yudis Fiction Prize.
External: WECode Technology Leadership Award (Harvard University) [Leadership and impact in STEM]; hackNY Fellowship (Columbia University & NYU) [fellowship for tech interns in NYC].

SKILLS

Technical	Python, Java, C/C++, Arduino Uno, Kubernetes, Docker, AWS, Snowflake, Redshift, PostgreSQL, Redis, Apache Kafka, Djangofly, RabbitMQ, Grafana, TSDB, Git, L ^A T _E X, Machine Learning, LLMs,
Languages	Hindi (Native/Bilingual), English (Fluent), Intermediate Sanskrit.